



EEG Spectra vs Recurrence Features in Understanding Cognitive Effort

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ABSTRACT

Characterizing cognitive states of a person based on surface electroencephalography (EEG) is traditionally performed by analyzing spectral power of the electrical brain activity at different frequency bands. Although frequency band activity over different electrode locations can provide valuable input for discriminating conscious and unconscious or drowsy and fully awoken brain states, it fails when fine grained cognitive state and effort estimation is required. This paper demonstrates that nonlinear features can be used to describe aspects of cognitive effort. Recurrence quantification analysis (RQA) features were used to discriminate mental relaxation, math calculations, and contemplating scientific articles. However, only when used in combination with traditional spectral features, the discrimination performance increases. The most dominant RQA features are recurrence rate and ratio, while the most dominant spectral features are relative beta and delta power.

CCS CONCEPTS

• **Applied computing** → **Health informatics.**

KEYWORDS

electroencephalography; EEG; recurrence quantification analysis; RQA; nonlinear features; cognitive effort.

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1 INTRODUCTION

Noninvasive electroencephalography (EEG) captures electrical activity stemming from synchronous firing of thousands to millions of neurons in the cerebral cortex. EEG is characterized as non-stationary or time-varying weak electrical signal, often compromised by various noise. Despite its non-linear nature, EEG signal analysis is mainly done in the linear domain, relying on the assumptions that EEG can be considered stationary over short time intervals and that noise-free segments can be isolated in a subset of the recorded channels. Using nonlinear analysis methods can overcome the pitfalls of linear analysis and provide an alternative and potentially better way of understanding brain activity [9].

The use of linear EEG analysis is rooted in the routine clinical practice used for almost a century. The main focus is on waveforms or spectral characteristics of EEG, addressing different frequency bands. Typical analysis of the conscious brain relies on estimations of the frequency band activity in the following bands: delta, theta, alpha, beta, and gamma. In less frequent investigations, statistical properties of the EEG are also studied, e.g., second and higher order statistics or signal extremes [14]. Often, statistical features are combined with spectral EEG features to derive useful information over the brain status.

Nonlinear methods have been used to describe changes in electrical brain activity preceding and during epileptic discharges and sleep stages. They rely on estimation of correlation dimension, entropy, nonlinear exponents, fractal dimensions, or recurrence patterns. The features derived based on these methods are shown to be at least as useful as linear features, particularly in epilepsy, sleep, and cognitive disorder research [15]. However, except few notable examples in the field of autism spectrum disorders (e.g., [7]) and cognitive impairments (e.g., [17]), they failed to reach wider clinical adoption. Non-straightforward interpretation and lack of support in clinical practice were among the main causes of this [9].

Combining linear and nonlinear EEG features has rarely been performed in practice, despite those being able to capture distinct aspects of EEG activity. Furthermore, using nonlinear features in wearable EEG applications that require real-time signal interpretation is atypical practice, mainly

due to computation complexity. However, with computational power of today's or future wearable devices and smart selection of source signals, e.g., limiting the number of EEG channels and features to use, this is a viable option [18].

In this paper we focus on understanding the contribution of linear and nonlinear features and investigating if their combining can bring benefits. We explore recurrence quantification analysis (RQA) features [10] as a representative of nonlinear features. Given that scalp EEG can only capture activity of a larger population of neurons, we focus on cognitive efforts for which we expect to engage a large neuronal population throughout the brain. Furthermore, the quasi-stationary assumption holds, assuming the absence of major brain state transitions and the use of short time intervals in the analysis. We investigate differences in the brain resting state activity, while it engages in mathematical operations, and while it contemplates scientific literature. The data is collected using a wearable EEG headset in an uncontrolled environment. As this resulted in a substantial contamination in the signal, the impact of rejecting noisy epochs from the analysis on the performance is also assessed.

2 METHOD

This section lists statistical, spectral and recurrence features used, followed by a description of the data source and data analysis methods.

Feature set

Statistical features. The set of statistical features includes: mean, standard deviation, maximum, minimum, median, mode, skewness, kurtosis, and zero crossing rate.

Spectral features. Typical frequency band based spectral features are derived, including delta (1–4Hz), theta (4–8Hz), alpha (8–13Hz), beta (13–20Hz), gamma (20–45Hz), and total (1–45Hz) band. Feature set includes absolute spectral powers as well as normalized ones (except for the total band), derived using Welch's method [19]. Features are normalized to the total band power. Based on a 1/f exponential EEG spectral model [12], spectral features also include intercept, slope, and fitting error, as well as individual alpha peak frequency (IAF), IAF power, and IAF width.

Recurrence features. Recurrence plots [4] capture when a phase space trajectory of a signal returns to a location it has visited before. Each element on a plot associates two points in time, in which the trajectory was at the same or sufficiently close place in a phase space. Hence, a recurrence plot contains specific structures such as single dots, diagonal and vertical lines, that can characterize an underlying system in terms of chaotic vs. deterministic nature of the signal. Recurrence quantification analysis (RQA) derives a number of features to characterize recurrence plots [10]. They capture

either density of recurrence points or the number and frequency of diagonal and vertical lines. In our analysis, we used the following 11 RQA features to quantify the recurrence plot of an EEG signals:

- Recurrence rate (**RR**) describes the density of recurrence points in a recurrence plot, i.e., the fraction of recurrence points over the total number of points.
- Determinism is the percentage of recurrence points which form diagonal lines.
- Laminarity is the percentage of recurrence points which form vertical lines.
- Ratio is the ratio of the percentage of recurrence points which form diagonal lines (determinism) and the percentage of recurrence points in the recurrence plot (RR).
- Averaged diagonal line length (**ADL**) is the average length of the diagonal lines in the recurrence plot.
- Trapping time (**TT**) is the average length of the vertical lines in the recurrence plot.
- Longest diagonal line (**LDL**) is the length of the longest diagonal line in the recurrence plot.
- Longest vertical line (**LVL**) is the length of the longest vertical line in the recurrence plot.
- Divergence is the inverse of the length of the longest diagonal line in the recurrence plot.
- Recurrence entropy is the Shannon entropy of the probability distribution of the diagonal line lengths.
- Trend is the paling of the recurrence plot towards its edges.

More details on the recurrence plot derivation and extraction of RQA features can be found elsewhere, e.g., [6]. The parameters required to estimate recurrence features, namely delay time, embedding dimension, and quantization threshold, are estimated based on average mutual information criterion [5], the false nearest neighbor algorithm [8], and empirical evaluation, respectively.

Database

The data used in this paper is collected using a wearable, wireless, dry-electrode EEG headset system with 4 active sensors that samples the data at 1024Hz [13]. The electrode locations were C3, C4, Cz, and Pz, according to the International 10-20 System for EEG recordings. Ten participants (seven male, between 23 and 37 years of age) participated in the experiment. Five different tasks were administered in a random order and performed in an uncontrolled setting, each task lasting for 3 minutes. Between each task a short break was applied (15-30s). The tasks were as follows:

- Eyes open (EO), mental relaxation task: participants had to relax having their eyes open.

- EO, mental calculation task: participants had to perform math calculations having their eyes open.
- EO, mental effort by reading task: participants had to read and contemplate on a scientific article of their choice having their eyes open.
- Eyes closed (EC), mental relaxation task: participant had to relax having their eyes closed.
- EC, mental calculation task: participant had to perform math calculations having their eyes closed.

The complete experiment consisted of three sessions per participant. Detailed description of the setup and protocol can be found in [12].

Data analysis

In the first step of data analysis, we band-pass filtered the signal in a 0.5 – 45Hz frequency band, and applied the 50Hz notch filter. We used a third order Butterworth filter. The Welch algorithm [19] was used to compute the spectra. Data was resampled to 128Hz and divided in 4s long epochs (i.e., time window) with no overlap per EEG channel. Each epoch was subjected to signal quality estimation algorithm [2]. Epochs assessed as having low signal quality were marked and either used or excluded from the analysis.

The features extracted included statistical, spectral, and recurrence features discussed above. To cope with the large variation in the feature values, standard (Z) scores are used. These scores normalize the feature values using the mean and standard deviation of feature values across different sessions, tasks, and participants. In the feature / performance analysis, the features set was reduced by removing features having correlation coefficient higher than 0.75, followed by a recursive feature elimination. Given the multi-class nature of the problem addressed and considering the feature analysis planned, we applied random forest to perform classification. A random sub-sampling 10-fold cross validation was used for performance evaluation. The classification was performed on the (1) two EC tasks, (2) three EO tasks and (3) all 5 EO & EC tasks. In addition, feature importance is investigated based on the random forest analysis in each of these classifications.

3 RESULTS

Signal quality

Application of signal quality indicator to remove noisy epochs resulted in halving the number of epochs in case of EO condition and reducing it by one third in case of EC condition, as shown in Table 1. The number of epochs removed per subject differs a lot (ranging from 7% to 85%), introducing a subject specific bias in the analysis of cleaned EEG data. The difference mainly stems from poor electrode-skin contact in some subjects and electrode locations, and due to the rigid headset design.

Table 1: The impact of removing low signal quality epochs.

Subject	EO			EC		
	Total	Clean	Removed	Total	Clean	Removed
1	480	290	39.6%	354	290	18.1%
2	413	319	22.8%	279	239	14.3%
3	427	312	26.9%	286	267	6.6%
4	416	244	41.4%	286	106	62.9%
5	430	217	49.5%	290	156	46.2%
6	421	113	73.2%	283	156	44.9%
7	417	186	55.4%	278	152	45.3%
8	415	297	28.4%	283	257	9.2%
9	421	109	74.1%	282	235	16.7%
10	448	65	85.5%	302	74	75.5%
Sum/Avg	4288	2152	49.8%	2923	1932	33.9%

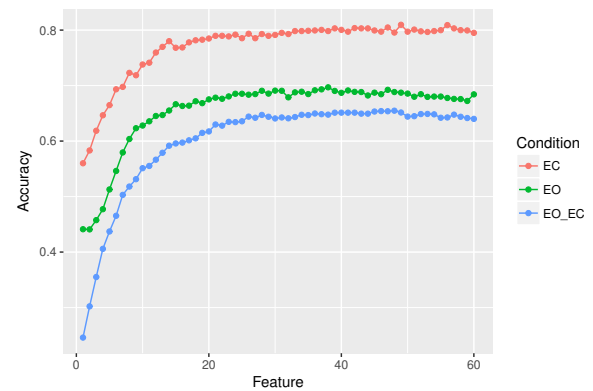


Figure 1: The classification performance with respect to the number of top ranked features included in the analysis.

Classification performance

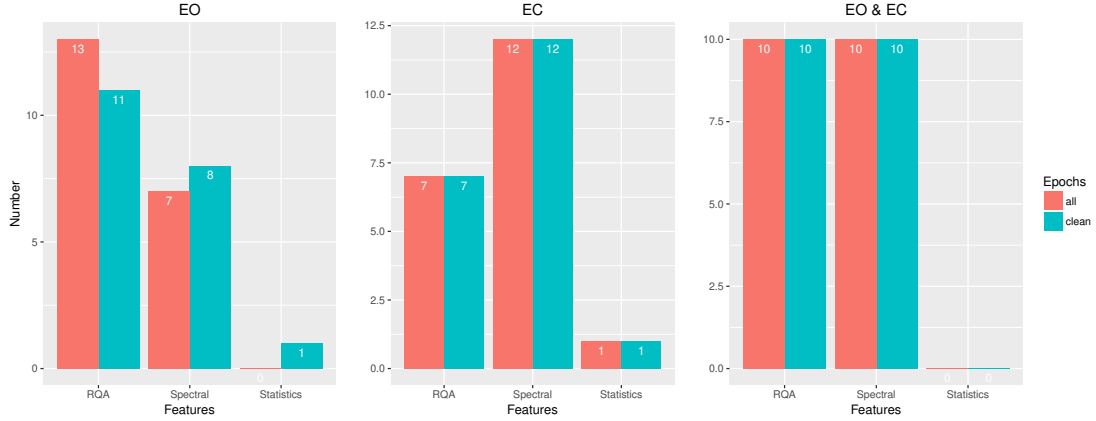
Table 2 indicates that spectral features solely can distinguish two tasks in the EC conditions quite well, while recurrence features are sub-optimal. Combining them neither increases accuracy nor Cohen's Kappa. However, in case of EO condition, though recurrence features have the same performance in artifact free data, when they are combined with spectral features, the performance increases compared to when only spectral features are used. The latter is also the case when distinguishing 5 classes in EO&EC conditions. The accuracy (Cohen's Kappa) when combining the two feature sets raises to 65% (.58) compared to 59% (.49) in case when using only spectral features and 53% (.42) when using recurrence ones. There is a minor impact of using only clean EEG epochs; overall, it slightly increases the accuracy when all features are combined and substantially only in case spectral features are used in EO and EO&EC tasks.

Feature relevance

The impact of top N features on the classification accuracy is shown in Figure 1. It is evident that to achieve stable performance at least top 20 features are required. Hence, in the analysis to follow, we explore top 20 features.

Table 2: Classification performance when using different sets of features: all (incl. RQA, spectral, and statistical), RQA, and spectral features on EO, EC, and combined EC & EC tasks.

Task	Epochs	Accuracy (%)			Cohen's Kappa		
		RQA&Spectral	RQA	Spectral	RQA&Spectral	RQA	Spectral
EO	all	69.1 ± 3.5	63.6 ± 2.2	63.7 ± 2.7	.536 ± .053	.454 ± .033	.456 ± .040
	clean	70.8 ± 3.5	62.8 ± 1.6	68.7 ± 1.5	.558 ± .052	.439 ± .023	.528 ± .023
EC	all	78.9 ± 2.6	71.2 ± 4.4	80.5 ± 3.0	.579 ± .051	.424 ± .088	.611 ± .061
	clean	80.1 ± 2.9	71.0 ± 3.8	80.7 ± 1.3	.603 ± .058	.419 ± .077	.614 ± .026
EO & EC	all	64.6 ± 1.6	53.3 ± 1.6	59.2 ± 1.1	.557 ± .020	.416 ± .020	.490 ± .014
	clean	65.5 ± 3.4	53.3 ± 2.5	62.6 ± 3.0	.565 ± .044	.410 ± .033	.528 ± .039

**Figure 2: The number of times different feature types are selected in across classification tasks.****Table 3: The number of times different features are present in the top 20 most important features across 10-folds.**

	EO				EC				EO & EC			
	all	#	clean	#	all	#	clean	#	all	#	clean	#
1	RR	4	β relative	3	δ relative	3	β relative	4	RR	4	RR	4
2	β relative	3	LDL	3	RR	3	δ relative	3	Ratio	4	Ratio	4
3	LDL	3	RR	3	β relative	3	RR	3	δ relative	3	δ relative	3
4	Ratio	3	Ratio	2	Ratio	3	Ratio	3	IAF power	2	IAF power	2
5	δ relative	2	δ relative	2	γ relative	1	γ relative	1	β relative	2	LDL	2
6	ADL	2	ADL	2	α absolute	1	θ relative	1	α absolute	1	α absolute	1
7	γ relative	1	Divergence	1	θ relative	1	β absolute	1	γ relative	1	γ relative	1
8	α absolute	1	γ relative	1	IAF power	1	ADL	1	IAF frequency	1	IAF frequency	1
9	Divergence	1	α absolute	1	ADL	1	α absolute	1	Divergence	1	β absolute	1
10			IAF power	1	β Absolute	1	IAF power	1	LDL	1	β relative	1

Although spectral features seem to have more dominant role in distinguishing cognitive efforts (see Table 2), this is not so obvious if we look at the top 20 most important features per classification task. Spectral features have higher importance only when distinguishing two EC conditions, same level of importance for EO&EC, and lower level of importance for EO conditions. This is illustrated in Figure 2. The importance of nonlinear features is even more evident when looking at the number of times different features are present within top 20 features per condition, shown in Table 3. Features occurring 3 or 4 times are predominantly RQA features, namely recurrence rate, longest diagonal, and ratio, having beta and delta relative features appearing twice or

twice. Using noise contaminated epochs has minor impact on the selection, slightly favoring spectral features.

4 DISCUSSION

Here we discussed the usefulness of spectral and recurrence features in characterizing cognitive effort based on EEG. The users performed three tasks involving different brain areas having their eyes either closed or open. Spectral features seem to achieve better performance based on the achieved accuracy levels compared to recurrence features. However, when exploring the number of features present in the top 20, it was evident that recurrence features play more dominant role. The most discriminating features were recurrence rate

and ratio, followed by relative power in delta and beta band. As indicated in Table 3, the former appeared 3 to 4 times in the list of most important features (maximum being 4). The limitation in terms of the number of EEG channels (4) could explain why RQA feature only classification performance was lower than spectral features only. Combining the two feature types resulted in increased accuracy, confirming that recurrence features capture important aspects complementary to spectral features when cognitive effort is in question. Although target population in this study was healthy users, these results suggest that combination of the two feature types can be applied to improve the performance of clinical approaches used to derive relevant information in ambulatory EEG. One obvious case is the detection of epileptic discharges, where usefulness of recurrence analysis is already demonstrated [1, 16].

One of the surprising outcomes of the analysis is that IAF based features were not among the important ones for discriminating cognitive effort, contrary to suggestions in the literature [11]. They were rendered as important only in few cases but well below dominant features. This effect could be attributed to interpersonal differences [12]. The observed changes in the beta frequency band are known to be related to alertness [3], but the change of relative delta activity is puzzling. One explanation could be the overall shift in the spectrum towards higher frequencies during cognitive tasks, given that delta band captures a large portion of total EEG spectral power.

Removing contaminated epochs from the analysis seem not to have substantial impact on the achieved performance, though it always resulted in positive trend when combining all or using spectral features. Interestingly, the performance slightly dropped when using only recurrence features, contrary to our expectation of RQA features being more sensitive to noise. To have better characterization of cognitive effort and investigate such outcomes, personalized analysis is required. It could provide insights on how these two feature types are enrolled in personalized models and explain the impact of those on the classification performance, with or without noisy epochs.

5 CONCLUSION

Nonlinear features derived from EEG, such as the ones based on recurrence quantification analysis, can bring complementary and potentially more valuable insights related to characterizing cognitive effort based only on spectral features. Recurrence rate and ratio have been the most useful recurrence features, irrespective of the cognitive effort and state being analyzed. Combining recurrence and spectral features can open new avenues for wearable ambulatory EEG monitoring such as predicting and detecting epileptic seizures but its usefulness is still to be determined.

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